

ELIO VINCIGUERRA

Catania, Italy
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EDUCATION

Ph.D. in Systems, Energy, Computer, and Telecommunications Engineering

University of Catania

Aug. 2023 - Jul. 2026

Catania, Italy

- Thesis: *Data-Driven Resilience in Next-Generation RISC-V Automotive Systems - From Microarchitectural Fault Injection to AI-Based Self-Healing.*

Master's Degree in Computer Engineering

University of Catania

Oct. 2021 - Jul. 2023

Catania, Italy

- Thesis: *Analysis and Optimization of Indoor Localization Algorithms for HPC Architectures*
- Final grade: 108/110

Bachelor's Degree in Computer Engineering

University of Catania

Oct. 2018 - Nov. 2021

Catania, Italy

- Thesis: *Evaluation of the Energy-Latency Trade-off in Deep Neural Network Accelerators*
- Final grade: 95/110

Diploma in Computer Science and Telecommunications

I.I.S.I.T. G. Marconi

Sept. 2013 - Jul. 2018

Catania, Italy

- Final grade: 94/100

EXPERIENCE

imec

gem5 Simulation R&D Engineer

Sept. 2026 - Present

Leuven, Belgium

- Extending the gem5 simulator with a system-level reliability and fault-impact assessment framework.
- Designing and implementing a fault injection framework into gem5.
- Integrating mini-applications and kernels for workload-aware analysis, correlating fault effects with application behavior and performance.

KTH Royal Institute of Technology

Visiting PhD Student

Apr. 2026 - May 2026

Stockholm, Sweden

- Use of the NoCDAS framework and integration of Transformers models and computational NoC paradigm within it.

KTH Royal Institute of Technology

Visiting PhD Student

Sept. 2025 - Jan. 2026

Stockholm, Sweden

- Use of the CGRA SiLago framework and integration of Deep Reinforcement Learning algorithms within it.

University of Catania

Teaching Assistance

Nov. 2023 - Jul. 2026

Catania, Italy

- Computer Architecture Course Teaching Assistance.
- Fundamentals of Computer Science Course Teaching Assistance.
- IoT Systems and Technologies Course Teaching Assistance.

Flazio S.r.l.

Intern

Nov. 2017 - Dec. 2017

Catania, Italy

- Web Design, website development using the Flazio platform.
- Social media integration.

BaxEnergy S.r.l

Intern

Jul. 2017 - Aug. 2017

Catania, Italy

- Quality control of company software programs.

Cultural Exchange Program at Polytechnic Institute Jin Yu

Sept. 2016

Catania, Italy

LANGUAGE PROFICIENCY

Italian	Native
English	Intermediate (B2)
Spanish	Intermediate (B1)
Chinese	Basic (A1)

TECHNICAL STRENGTHS

Programming Languages	Python, C, C++, Java, Go, C#
Web Technologies	HTML, CSS, PHP
Databases	SQL (MySQL, PostgreSQL, Microsoft SQL)
Tools	Git, VS Code, IntelliJ, Eclipse

PUBLICATIONS

1. E. Russo, F. G. Blanco, E. Vinciguerra, D. Patti, G. Ascia and M. Palesi, "COSMA: Communication aware Optimization of Fermionic Simulation Kernels for Modular Quantum Architectures," accepted at *IEEE International Conference on Quantum Computing and Engineering (QCE)*, Sept. 2026. arXiv: 2607.09381. [Online]. Available: <https://doi.org/10.48550/arXiv.2607.09381>.
2. E. Vinciguerra, N. Nosrati, G. Ascia, M. Palesi, and A. Hemani, "SPACE: Safety-critical Performance-counter Adaptive Control Engine for Anomaly Mitigation via Deep Reinforcement Learning," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2026, doi: 10.1109/TVLSI.2026.3722739.
3. E. Vinciguerra, G. Siracusano, G. Ascia and M. Palesi, "SAFER: Self-Attentive Feature Extraction and Reduction," in *IEEE Access*, vol. 14, pp. 85695-85708, 2026, doi: 10.1109/ACCESS.2026.3700810.
4. E. Vinciguerra, E. Russo, G. Ascia, and M. Palesi, "CHAOS: Controlled hardware fault injector system for gem5," 2026. arXiv: 2602.02119 [cs.AR]. [Online]. Available: <https://arxiv.org/abs/2602.02119>.
5. E. Vinciguerra, M. Palesi, and G. Ascia, "Improving Machine Learning Anomaly Detection for Safety-Critical RISC-V Automotive Systems," in *Proceedings of the CPS Summer School PhD Workshop 2025*, C. Pilato et al., Eds., vol. 4106. Aachen, Germany: CEUR-WS.org, 2025. [Online]. Available: <https://ceur-ws.org/Vol-4106/short1.pdf>.
6. E. Russo, E. Vinciguerra, M. Palesi, D. Patti, G. Ascia and V. Catania, "TeleSABRE: Heuristic Layout Synthesis in Multi-Core Quantum Systems with Teleport Interconnect," 2025 *IEEE International Conference on Quantum*

Computing and Engineering (QCE), Albuquerque, NM, USA, 2025, pp. 749-758, doi: 10.1109/QCE65121.2025.00086.

7. E. Vinciguerra, E. Russo, M. Palesi, G. Ascia, "An Anomaly Detection Model for RISC-V in Automotive Applications: A Domain-Specific Accelerator Perspective," 2025 33rd Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP), Turin, Italy, 2025, pp. 317-320, doi: 10.1109/PDP66500.2025.00051.
8. E. Vinciguerra, E. Russo, M. Palesi and G. Ascia, "Data-Driven Simulation Based Fault Detection in Automotive RISC-V Applications," 2024 *IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc)*, Kuala Lumpur, Malaysia, 2024, pp. 509-516, doi: 10.1109/MCSoc64144.2024.00089.
9. E. Russo, E. Vinciguerra, G. Ascia and D. Panno, "DNN Hardware Accelerator Selection for Feasible Deployment of MARL-Based MAC Protocols in Industrial IoT Networks," 2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI), Milano, Italy, 2024, pp. 443-448, doi: 10.1109/RTSI61910.2024.10761172.
10. E. Vinciguerra, E. Russo, M. Palesi, G. Ascia and H. Rafique, "Improving LSTM-based Indoor Positioning via Simulation-Augmented Geomagnetic Field Dataset," 2024 *IEEE International Conference on Mobility, Operations, Services and Technologies (MOST)*, Dallas, TX, USA, 2024, pp. 251-259, doi: 10.1109/MOST60774.2024.00034.